SQL Exercise : Aggregate functions and Grouping

Questions

1. **COUNT function** - write a SQL query to find the total number of employees in the company

→ SELECT COUNT (DISTINCT ID) AS NO_of_employees
FROM employees;

NO_of_employees
10

2 **SUM function** - write a SQL query to find the total Salary paid to all employee in the IT department

→ SELECT SUM(Salary) AS total_Salary
FROM employees
WHERE department = 'IT';

total_Salary
543 000

3. **AVG function** - write a SQL Query to calculate the Average Salary of employees in the HR department

→ SELECT AVG (Salary) AS Average_Salary
FROM employees
WHERE department = 'HR';

Average_Salary
49 500

4, <u>MIN and MAX function</u> - write a SQL query to find the highest and lowest Salary in the company

```
-> SELECT MIN (Salary) AS Lowest_Salary
          MAX (Salary) AS Highest_Salary
   FROM employee
```

Lowest_Salary	Highest_Salary
48 000	62 000

5, <u>GROUP BY Statement</u> - write a SQL Query to group employees by department and display the total Salary paid in Each department

```
-> SELECT department,
          SUM (Salary) AS total_Salary
   FROM employees
   GROUP BY department;
```

Department	Total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6, <u>GROUP BY and COUNT</u> - write a SQL query to Count how many employees work in each city

```
-> SELECT city,
          COUNT (id) AS NO_of_employees
   FROM employees
   GROUP BY city;
```

City	No_of_employees
New york	2
Chicago	3
Los Angels	2
San fransiso	2
Houston	1

7. <u>GROUP BY and ORDER BY</u> - write a SQL query to group employees by department, calculate the average salary in each department and order the results in descending order of average salary

```
→ SELECT department
          AVG (Salary) AS Average_Salary
  FROM employees
  GROUP BY department
  ORDER BY Average Salary DESC;
```

Department	Average_Salary
Finance	59 500
IT	5 500
Marketing	52 500
HR	49 500

8 <u>HAVING Clause</u> - write a SQL query to find department where the total salary paid exceeds 100 000

```
→ SELECT department,
          SUM (Salary) AS total_Salary
  FROM employees
  GROUP BY department
  HAVING SUM (Salary) > 100 000;
```

Department	total_Salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY - write a SQL Query to list cities where more than one employee works, ordered by the number of employees in descending order

```
-> SELECT City,
          COUNT(id) AS NO_of_employees
   FROM employees
   GROUP BY City
   HAVING COUNT(id)>1
   ORDER BY count(id) DESC;
```

City	NO_of_employees
Chicago	3
New york	2
Los Angels	2
San francisco	2

10 Combining Aggregate functions - write a SQL query to find the department with the highest average Salary

```
-> SELECT department,
          Avg(Salary) AS Avg_Salary
   FROM employees
   GROUP BY department
   ORDER BY Avg_Salary DESC;
```

department	Avg_salary
Marketing	70000